

BHABHA UNIVERSITY, BHOPAL

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BACHELOR OF COMPUTER APPLICATIONS (BCA)



**DETAILED SYLLABUS
(EFFECTIVE FROM JULY 2018)**

**Department of Computer Science & Application
BHABHA UNIVERSITY, BHOPAL**

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DETAILED SYLLABUS

S.No.	Paper Code	Paper Name	Internal Evaluation	Theory	Practical	Total
1	BCA21	Operating System	25	50		75
2	BCA22	Data Structure	25	50		75
3	BCA23	DBMS	25	50		75
4	BCA24	Computer Oriented Numerical Analysis	25	50		75
5	BCA25	C++	25	50		75
6	BCA26	Software Engineering	25	50		75
Practical						
7	BCA27	DBMS with SQL	20		30	50
8	BCA28	Lab C++	50		50	100
Total			220	300	80	600

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BCA21- OPERATING SYSTEM

UNIT I

Operating System Definitions, its Components, Evolution of Operating System, types of operating systems: batch, multi programmed, multitasking, desktop, multiprocessor, real-time, client-server, peer-to-peer, distributed, clustered and handheld. Operating system services, dual-mode operation ,protection of I/O, memory and CPU. Non-virtual and virtual machines.

UNIT II

Process Scheduling: concept of a process, process states, PCB, Process state transitions, operations on processes, context switch, types of schedulers, CPU scheduling concept, CPU Scheduler, CPU-I/O burst cycle, dispatcher, scheduling criteria, scheduling algorithms – FCFS, SJF, STRN, Round Robin, priority, multilevel queue and event driven (i.e., pre-emptive priority) scheduling algorithms, Performance evaluation of algorithms through deterministic modeling.

UNIT III

Memory Management Concepts: Address Binding, logical and physical address space, dynamic loading etc., Contiguous allocation methods – static & dynamic partitioned memory allocation. Concepts of fragmentation, swapping, relocation, compaction, protection, sharing. Segmentation. Non-contiguous allocation methods – Paging: basic principle of operation, h/w support for paging, protection and sharing. Virtual memory: concept of demand paging, Page fault, page replacement algorithms – FIFO, LRU, OPT. Thrashing, Concept of Page fault frequency, prepaging , decision about minimum number of frames, page size etc.,

UNIT IV

File system implementation, Responsibilities of file management system, directory implementation as linear list/hash table, directory structure, disk organization, disk controller and driver, disk space management – contiguous allocation, non contiguous allocation – chaining and indexing, disk address translation. Idea about disk caching, disk mirroring. Disk scheduling algorithms. Disk management.

UNIT V

Device Management: I/O hardware, Techniques for device management. Dedicated devices, shared devices, virtual peripherals. Security & protection: Security threats and goals, penetration attempts, Security policies and mechanism, authentication, protection and access control. Interprocess communication, need for interprocess synchronization. Deadlocks – definition, avoidance, detection, prevention and recovery.

TEXT BOOK: 1. Operating System Concepts – by Silberschatz, Galvin and Gagne.
2. Operating System Concepts and Design – by Milenkovic
3.. Operating System – by Tanenbaum.
4. Operating System – by Peterson

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BCA22- Data Structure

UNIT I

Definition of Data Structure, its types, Basic operations in Data Structures. Array – its representation, Operations on arrays, address calculation of single and multi dimensional arrays in memory, Applications of arrays.

UNIT II

Stack: Representation of Stacks, operations on Stacks, infix and postfix notations, postfix expression evaluation, application of Stacks. Queues: Representation of queues, operations on queues, circular queue, Dequeue, Priority Queue, Applications of queues.

UNIT III

Linked list: Single Linked List – description & operations, Doubly Linked List – description & operations, Linked implementation of Stacks and queues, Header linked list.

UNIT IV

Trees: Basic terminology, binary tree, binary search tree, Operations on binary tree: Insertion & Deletion algorithms. Traversal of binary trees: Inorder, Preorder & Post order.

UNIT V

Searching: Linear Search, Binary Search. Sorting: Selection Sort, Bubble Sort method & its algorithm, Quick Sort, Merge Sort. Graphs: Related Concepts and its representations. Graph Traversal Schemes: Depth first search, Breadth first search.

TEXT BOOKS:

1. Data Structure by Schaum Series.
2. Data Structure by Tanenbaum.
3. Data Structure by Sartaj Sahani
4. Data Structure using C++ by Yashwant Kanetkar

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BCA23- Database management system

UNIT 1

Basic Concepts of Data File system, its advantages and disadvantages. Concepts of DBMS: Data, Information, Database, Components of DBMS, Architecture of a database system – Physical, Conceptual and User level, Data Independence – Logical and Physical, DBMS terminology, Data Dictionary. Database Models: Network, Hierarchical and Relational Models, Features and Comparison of the three models. Concepts of Multitier Architecture in databases, Brief idea about the new concepts like distributed databases, parallel databases, mobile databases, temporal databases, spatial databases, geographic databases, data warehousing & data mining.

UNIT 2

RDBMS: Definition, Components, Terminology, Difference with DBMS. ER-Model, ER-Diagram, ER-concepts, types of relationships. Codd's 12 rules, Normalization: Introduction, concepts associated with normalization, key terminology: primary, candidate, foreign, alternate keys. Steps involved in normalization, 1st, 2nd, 3rd, 4th normal forms, BCNF, 5th normal form.

UNIT 3

Idea about Generalization, Aggregation, Specialization. Relational Algebra: Formal Definition, Fundamental Operations – select, project, union, set, difference, Cartesian product & rename, additional operations & extended operations. Concept of SQL sublanguages – DDL, DML, DCL, TCL, SCL etc., Embedded SQL. Interactive SQL: Invoking sql*plus. Oracle data types, two dimensional matrix creation, Modifying the structure of tables, dropping tables.

UNIT 4

DML commands: Insertion, updation, deletion operations, many faces of select command, data constraints, logical operators, range searching, pattern matching, oracle functions, use of Alias, grouping data from tables, manipulating dates in sql, joins, Sub-queries, indexes, views, sequences, roles, synonyms, use of save point, rollback, commit commands, creating user accounts, granting permissions, revoking permissions. Concept of importing and exporting database files.

UNIT 5

SQL: Introduction, the SQL execution environment, the SQL syntax, block structure – declarative part, executable part, exception handling part, variable declaration using type, row type, if statements, looping structures, oracle transactions, cursors & its types, cursor attributes, nesting of cursors, parameterized cursors, error handling in SQL, locks.

REFERENCES BOOKS:

1. Database System Concepts – by Silberschatz, Korth, Sudarshan (Mc Graw Hill).
2. An Introduction to Database Systems – by Bipin Desai.
3. SQL, PL/SQL – The Programming language of Oracle – by Ivan Bayross (BPB).
4. Introduction to Database Systems by C J Date (Pearsons Education).
5. SQL/PLSQL for Oracle by P.S.Deshpande (AWI).

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BCA24- Computer Oriented Numerical Analysis

UNIT I

Solutions of Algebraic, Transcendental & Simultaneous Algebraic Equations. Introduction to Algebraic & Transcendental Equations, Newton-Raphson method, Regula-Falsi method, Successive Bisection method, Secant method, Comparison of Regula-Falsi method with Bisection method, Gauss Elimination method, Gause-Seidal method, Gauss Jordan method, Concept of Pivoting.

UNIT II

Interpolation & Extrapolation: Newton's Interpolation formulae, Newton's forward difference & Backward difference formula, Lagrange's Interpolation formula, Newton's divided-difference interpolation, Sterling's and Bessel's Central difference formula.

UNIT III

Numerical Integration & Differentiation: Introduction, Newton's Cotes formulae, Trapezoidal rule, Simpson's rule, Weddle's rule.

UNIT IV

Solution of ordinary differential & Integral Methods: Euler's, Picard's and Taylor's series methods. Modified Euler method, Picard's method for successive approximation, method of successive differentiation, Runge-kutta second & forth orders methods.

UNIT V

Correlation & Regression: Karl Pearson's Coefficient of Correlation, Rank Coefficient of Correlation, Partial & Multiple Correlation, Lines of Regression, Coefficient of Regression.

REFERENCES BOOKS:

1. Method of Numerical Methods by Shastri.
2. Computer Based Numerical Algorithms by Krishnamurthy.
3. Computer Oriented Numerical Methods by Salvadori.
4. Numerical Methods by H C Agarwal.
5. Fundamentals of Mathematical Statistics by Gupta & Kapoor.

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BCA25- C++

UNIT I

Principles of OOP, procedure oriented programming vs. object oriented programming, basic concepts, advantages, application of OOPs, object oriented languages. Beginning with C++: What is C++, structure of C++ program, creating, compiling, linking & executing a C++ program, Tokens, expressions & control structures, keywords, identifiers, basic data types, user-defined data types, derived data types, symbolic constants, type compatibility, variable declaration, dynamic initialization of variables, reference variables.

UNIT II

Operators in C++: scope resolution operator, memory management operators, manipulators, type cast operators, operators, operator precedence, control structures. Functions in C++: Main function, function prototyping, call by reference vs. call by value, inline functions, default arguments, const arguments, function overloading, friend functions. Classes and objects: specifying a class, defining member functions, making an outside functions inline, private member function; array within a class, memory allocation for object; static data members, static member functions, array of objects, objects as function arguments, returning objects.

UNIT III

Constructors and Destructors: Constructors, Parametric Constructors, Multiple Constructors in a class, constructors with default arguments. Dynamic initialization of objects, copy constructors, dynamic constructors, destructors. Operator Overloading & Type Conversions: Definition of Overloading, & Operator Overloading, rules for Overloading Operators, Overloading Unary Operators, Binary Operators, Binary Operators using Friends.

UNIT IV

Inheritance: defining derived classes, single inheritance, multilevel inheritance, multiple inheritance, hierarchical inheritance, hybrid inheritance, virtual base class, abstract classes, constructors in derived classes, member class, nesting of class.

UNIT V

Pointers, virtual functions and polymorphism, pointers to objects, this pointer, pointers to derived class, pure virtual functions, exception handling in C++, managing console I/O operations, working with files :open, close, basic read-write operations on files .

TEXT BOOKS:

1. Object Oriented Programming with C++ by E Balagurusamy.
2. Programming in C++ by Robert Lafore
3. C++ - The complete Reference – by Herbert Schildt (TMH)
4. Programming with C++, Schaum Series
5. OOP's concepts – by David Parsson

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BCA26- Software Engineering

UNIT I

Introduction of system concept, characteristics of system, elements of system, The System Development Life Cycle, The Role of System Analyst. Introduction system planning & initial investigation, various information gathering tools feasibility study conretions & structures tools of system analysis, various methods of process design, form design methodologies

UNIT II

Software Process, Product and Project: The Product : Software, Software Myths, The process : Software Engineering : A Layered Technology, Software Process Models, The Linear Sequential Model, The Prototyping Model, The RAD Model, Evolutionary Software Process Models

UNIT III

Software Project Planning and Design : Software Project Planning : Project planning objectives, Decomposition Techniques, Empirical estimation models, The Make/Buy Decision., Risk analysis. Software Design: Design Principles, Cohesion & Coupling, Design notation and specification, structure design methodology.

UNIT IV

Software Quality Assurance and Testing: Software Quality Assurance : Quality Concepts, The Quality Movement, Software Quality Assurance, Software Reviews, Formal Technical Reviews, Formal Approaches to SQA, Statistical Software Quality Assurance, Software Reliability, Mistake Proofing for Software

UNIT V

Advanced Topics: MIS & DSS: Introduction to MIS, long range planning, development and implementation of an MIS, applications of MIS in manufacturing sector and in service sector. Decision Support System concepts, types of DSS. Object Oriented Software Engineering: Object Oriented Concepts, Identifying the Elements of an Object Model, Management of Object Oriented Software Projects. CASE tools, Re-engineering

BOOKS

1. Richard Fairley : Software Engineering Concepts (TMH)
2. Pankaj Jalote “Software Engg” Narosa Publications.
2. Ian Sommerville : Software Engineering 6/e (Addison-Wesley)
3. Richard Fairley : Software Engineering Concepts (TMH).